# Check of your `BFKL.tex`

Notation as in your file: $\mathcal{K}(X) = X/X^2$, $U^{bc} = U^{cb}$. Blue: new or corrected; red: removed. Reference: the log $\frac{\vee}{k^+}$ list of every row in `Checks_JIMWLK_BFKL.pdf` (eq. (B:...)), which comes from the exact split of your current Group I–III rows, and BFKL⊗LO of App. E (Eq. (LOBFKL)).

**Method.** Every equation is evaluated pointwise: random transverse points, random classical charges, $U = U^T$, $\log \frac{\vee}{k^+} = 1$, and $\epsilon = 0.37$ kept finite in the pole brackets.

- Three-$\rho$ first forms are compared with the reference after symmetrization over the charges (with the c.c. for Groups I, III).
- The "This can be written as" lists are compared with the two-$\rho$ (commutator) terms of the first form, computed directly. With the charges as operators, a product $ABC$ of three charges is its symmetric part plus $\frac{1}{2}([A,B]C+[B,C]A+[A,C]B)$, with $[\rho^a(x), \rho^b(y)] = if^{abc}\rho^c(x)\,\delta^{(2)}(x-y)$; anticommutators are kept ordered, $\{A,B\} = AB+BA$, and the c.c. reverses the order.
- This rule was checked in two ways. It reproduces the reordering engine used before term by term (6 rows, $\leq 2\times10^{-16}$). It also reproduces the checked two-$\rho$ lists of `Evolution_terms.tex` from the log $\frac{k^+}{\Lambda}$ three-$\rho$ terms (Group II Row I, Group III Rows I, II: exact).
- Index sums ($\delta^{ij}$, $(x-w)^i\ldots$) are done term by term. A term without the index is not summed over it.

## 1 Answer

- **Your result is correct.**
  - The sum of all two-$\rho$ lists of the file is BFKL⊗LO exactly ($1.2\times10^{-15}$, two independent evaluators).
  - Both forms of your "BFKL contribution" are BFKL⊗LO ($2.2\times10^{-15}$), and they equal that sum.
  - The colour-singlet form is unchanged from the version checked before; it equals Eq. (LOBFKL) of App. E.
- **Row by row the file is correct, with three corrections (C1–C3):**
  - every first form equals the log $\frac{\vee}{k^+}$ terms of its row (Table 1), except Group I Row II (C1);
  - every "written as" list is exactly the two-$\rho$ part of its first form (Table 2), Group I Row II with the corrected first form;
  - the two-$\rho$ lists "From 4$\rho$'s" are identical to those of `Evolution_terms.tex` (difference 0).

## 2 Corrections

**C1 Group I, Row II, first form.** The equation as written is a copy of Group I Row III, $d^3N_1$ (identical text). The log $\frac{\vee}{k^+}$ terms of Row II are (`Checks_JIMWLK_BFKL.pdf`, eq. (B:G1II2a)):

$$\begin{aligned}
\frac{d^3N_{2\,a}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\\
&\times\left[\frac{(x-w)^i}{(x-w)^2}\,\frac{(w-z)\cdot(y-z)}{(w-z)^2\,(y-z)^2}-\frac{(y-w)^i}{(y-w)^2}\,\frac{(w-z)\cdot(x-z)}{(w-z)^2\,(x-z)^2}\right]\\
&\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^4}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x'}\frac{(x'-w')^i}{(x'-w')^2}\\
&\times\left[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2}-\frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\right]\\
&\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big]. \qquad (1)
\end{aligned}$$

The logarithmic term has no $z$ left, so its measure is $\int_{x,y,x'}$. Your "This can be written as" list for Row II is exactly the two-$\rho$ part of these terms ($1.9\times10^{-15}$), so only the displayed first form changes.

**C2 Group II, Row IV, third line** ($H_4$): the sign at the start of the line is missing; it is $+$. Without it LaTeX prints the line as a product. It was also missing in my `Checks_JIMWLK_BFKL.pdf`; corrected there.

**C3 Colour-singlet form:** the measure is $\int_{x,y,z,w,w'}$ ($y$ is used, there is no $x'$).

# 3 Row by row

| your row (first form) | reference | relative difference |
|---|---|---|
| Group I, Row II | (B:G1II2a) | different (C1); corrected: 0 |
| Group I, Row III, $d^3N_1$, $d^3N_2$ | (B:G1IIIN1), (B:G1IIIN2) | 0, 0 |
| Group I, Row IV | (B:G1IV) | 0 |
| Group II, Row I, first part (two equations) | (B:G2I1A), (B:G2I1B) | 0, 0 |
| Group II, Row I, second part (five equations) | (B:G2I2b)–(B:G2I2f) | 0 |
| Group II, Row IV | (B:G2IV) | 0 (with C2) |
| Group III, Rows I, II, III (both parts), V, VI | (B:G3I), ..., (B:G3VI) | 0 |

Table 1: First forms against the log $\frac{\vee}{k^+}$ lists, pointwise (with the corrected index sums).

| row | "written as" list against | relative difference |
|---|---|---|
| Group I, Row II | two-$\rho$ part of the corrected first form | $1.9 \times 10^{-15}$ |
| Group I, Row III | two-$\rho$ part of $d^3N_1 + d^3N_2$ | $1.0 \times 10^{-15}$ |
| Group I, Row IV | two-$\rho$ part of the first form | $6.0 \times 10^{-15}$ |
| Group II, Row I, first part | two-$\rho$ part of the first form | $1.8 \times 10^{-14}$ |
| Group III, Row I | two-$\rho$ part of the first form | $4.4 \times 10^{-15}$ |
| Group III, Row II | two-$\rho$ part of the first form | $2.9 \times 10^{-15}$ |
| Group III, Row III, first part | two-$\rho$ part of the first form | $7.5 \times 10^{-16}$ |
| Group III, Row V | two-$\rho$ part of the first form | $4.4 \times 10^{-16}$ |
| Group III, Row VI | two-$\rho$ part of the first form | $4.8 \times 10^{-14}$ |
| Group II, Row I, second part | the five first-form equations | $1.9 \times 10^{-14}$ |
| Group II, Row IV | the first form (the first two lines cancel; $H_4$ stays) | $8.8 \times 10^{-14}$ |
| Group III, Row III, second part | the first form (with its c.c.) | $1.6 \times 10^{-14}$ |
| From 4$\rho$'s, $d^3N_1$, $d^3N_2$ | the same lists in `Evolution_terms.tex` | 0, 0 |
| sum of all two-$\rho$ lists | BFKL⊗LO | $1.2 \times 10^{-15}$ |
| BFKL contribution, first / collected form | BFKL⊗LO | $2.2 \times 10^{-15}$ / $2.2 \times 10^{-15}$ |

Table 2: The "written as" lists. Groups I and III include their c.c. (your lists are written with it).

# 4 Reproducing

In `nlo_gluon_paper/bfkl_tex_check/`:

```
python3 rerun_first.py      # first forms vs log(vee/k+) lists -> cmp_first.json
python3 cmp_two_rows.py     # two-rho rows: written vs first -> cmp_two_rows.json
python3 run_reorder.py      # three-rho rows: written vs two-rho part -> cmp_reorder.json
python3 rerun_rows.py       # G1 IV, G2 I, G3 I, G3 II  -> cmp_reorder_fixed.json
python3 two_total.py        # all two-rho lists vs BFKL x LO -> two_total.json
python3 rule_vs_engine.py   # ordering rule vs the engine   -> rule_vs_engine.json
python3 test_tool.py; python3 test_tool2.py   # rule on the J terms
python3 make_corrected.py; python3 make_corrected.py marked   # corrected/
```

`reorder.py` computes the two-$\rho$ part; `texfix.py` does the index sums term by term.